Ge 158 Physical Principles of Remote Sensing
Problem Set 1: Dielectric properties, EM waves, and reflection at normal incidence

Notation and conventions (as in the lecture notes): Time-harmonic fields use $e^{j\omega t}$ with $j^2 = -1$, so the complex dielectric constant is $\varepsilon = \varepsilon' - j\varepsilon''$ with $\varepsilon'' \geq 0$. A tilde denotes a phasor, $\mathbf{E}(x, y, z, t) = \mathrm{Re}[\tilde{\mathbf{E}}(x, y, z)\, e^{j\omega t}]$. Use $c = 3 \times 10^8$ m/s, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

Problem 1: The magnetic field of a plane wave propagating in a lossless, homogeneous, isotropic, nonmagnetic medium is given as

$$\mathbf{H} = 10 \cos\left(2\pi \times 10^9 t - 20\pi y\right)\hat{\mathbf{x}} \qquad (\mathrm{mA/m}) \tag{1}$$

where t is in seconds and y is in meters. Determine the (a) direction of propagation, (b) phase velocity u_p, (c) wavelength λ, (d) relative permittivity ε' of the material, and (e) the electric field phasor $\tilde{\mathbf{E}}$.

Problem 2: Consider a uniform plane wave propagating in the $+x$ direction through a lossless, nonmagnetic medium with relative permittivity $\varepsilon' = 4$. The electric field has a frequency of 10 GHz, a peak value of 12 V/m, and is oriented along $\hat{\mathbf{z}}$. At $t = 0$ and $x = 0.02$ m, the instantaneous value of E_z is measured to be 6 V/m and is decreasing in time. Give expressions for the electric and magnetic fields of the form $\mathbf{E}(x, t) = |E_z^+| \cos(\omega t - kx + \phi^+)\hat{\mathbf{z}}$, where ϕ^+ is the phase, and similarly for $\mathbf{H}(x, t)$ (be careful with its direction).

Problem 3: For this problem, use the Debye relaxation model given below. Water is a lossy medium, so use the exact expressions for the attenuation and phase constants (Lecture 2), not the low-loss approximations.

a. Calculate the skin depth δ_s, penetration depth δ_p, and phase velocity u_p for an EM wave with the same frequency (in Hz) as the wave given in Problem 1 propagating through pure water at temperatures $T = 10$ °C and $T = 0$ °C.

b. Now suppose that the frequency of the wave is multiplied by a factor of 20. How do δ_s, δ_p, and u_p for water at $T = 0$ °C change?

c. Now suppose the water is slightly brackish, with an electrical conductivity of $\sigma = 0.5$ S/m. As a rough approximation (the double-Debye model from Lecture 1 is the proper treatment of salt water), represent the effect of the dissolved salt by adding the conduction term $\sigma/(\omega\varepsilon_0)$ (Lecture 2) to ε'' from the Debye model. Recalculate δ_s, δ_p, and u_p for the cases in parts (a) and (b). Then double the conductivity to $\sigma = 1$ S/m and repeat. At which frequency does conductivity matter more, and why?

Debye model (modified from Ulaby and Long, 2014, Chapter 4):

$$\varepsilon = \varepsilon_{w\infty} + \frac{\varepsilon_{w0} - \varepsilon_{w\infty}}{1 + j\omega\tau_w} \tag{2}$$

where $\varepsilon_{w\infty} = 4.9$,

$$\tau_w = \frac{1}{2\pi}\left(1.11 \times 10^{-10} - 3.82 \times 10^{-12}T + 6.94 \times 10^{-14}T^2 - 5.1 \times 10^{-16}T^3\right) \tag{3}$$

and

$$\varepsilon_{w0} = 88.05 - 0.41T + 6.3 \times 10^{-4}T^2 + 1.08 \times 10^{-5}T^3 \tag{4}$$

All temperatures should be given in °C, frequencies in Hz, and τ_w comes out in seconds.

Problem 4: A radar looking straight down (normal incidence) from the air ($\varepsilon_1 = 1$) observes smooth, flat surfaces at frequencies of 1 GHz and 20 GHz. Consider three surfaces (medium 2): (i) pure water at $T = 0$ °C (use your values of ε from Problem 3); (ii) brackish water at $T = 0$ °C with $\sigma = 0.5$ S/m (use your values of ε from Problem 3c); and (iii) freshwater ice, with $\varepsilon \approx 3.2$ at both frequencies (Lecture 1). For the ice, consider only the reflection at the air–ice interface (we will treat layered media in Lecture 6).

a. For each surface and each frequency, calculate the Fresnel reflection coefficient ρ (give its magnitude and phase), the reflectivity $\Gamma = |\rho|^2$, and the transmissivity $\mathbb{T} = 1 - \Gamma$. Organize your results in a table.

b. Using your table, which surface appears brightest to the radar, and which appears darkest? Does the brackish water look appreciably different from the pure water? Explain your answers in terms of the dielectric properties of each medium.

c. Using your table, is the reflected wave in phase or out of phase with the incident wave in each case? For which cases is the phase offset exactly π, and why do the others depart from π?

d. *Water only (surfaces i and ii)*: Use the penetration depth δ_p from Problem 3 to calculate the depth below the water surface at which the power of the transmitted wave has fallen to 1% of the power of the incident wave, at each frequency. Account for the power reflected at the surface. How do your answers compare between frequencies?

e. *Ice only (surface iii)*: Now consider a wave traveling upward in ice that is normally incident on the ice–air interface (as for a radar sounding a glacier from within the ice). Calculate ρ and compare it with your result for the air–ice interface. Is the reflected wave in phase or out of phase with the incident wave?